Project: Worked examples
Title: Enclosure heat dissipation and fan sizing
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

## 1. Scope and method

Total heat load inside a control enclosure, the effective surface that can dissipate it for the mounting condition, the internal temperature rise with natural convection and radiation through a stated overall heat-transfer coefficient, the forced-ventilation airflow for a target rise (density corrected for altitude), the fan duty point against a fan curve and the system resistance, optional solar gain, and the cooling-unit capacity needed when the limit cannot be met.

| WHAT IT CALCULATES | RELATION |
|---|---|
| Heat dissipated inside the enclosure, plus any solar gain | $P = \Sigma P_i n_i d_i (+ \alpha I A_{sun})$ |
| Effective cooling surface, each face weighted by how freely it sheds heat | $A_{eff} = \Sigma f_{face} \times A_{face}$ |
| Temperature rise with natural cooling, one coefficient over the exposed faces (the IEC TR 60890 tables are not used) | $\Delta T = P/(h A_{eff})$ |
| Air flow a fan must move to hold the target rise | $V = P/(\rho c_p \Delta T_{target})$, with $\rho = p(z)/(287.055 T)$ and $c_p = 1006$ J/(kg·K) |
| Fan duty point, where the fan curve meets the system resistance (found by bisection) | $p_{fan}(V) = p_{sys} (V/V_{ref})^2$ |
| Cooling unit capacity, when the surfaces alone cannot reject the heat | $P_{cool} = P - h A_{eff} (T_{max} - T_{amb})$ |

Basis: A method of temperature-rise verification of low-voltage switchgear and controlgear assemblies by calculation[1]; Effective surface, heat-transfer coefficient and fan-sizing practice (cited, not copied)[2]; Verification of temperature rise[3](10.10).

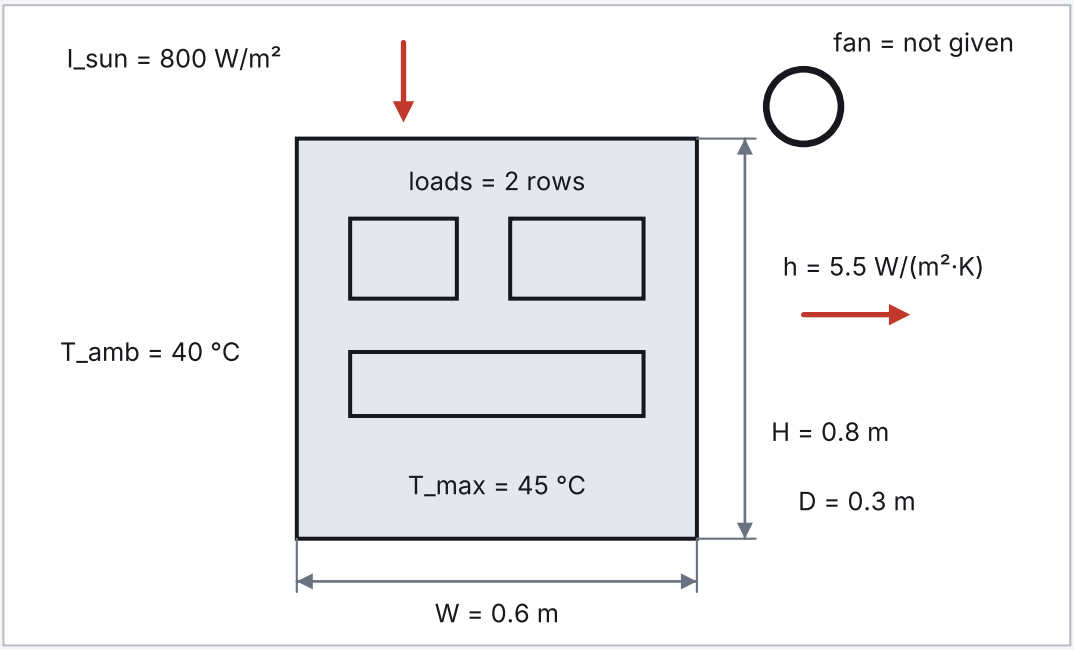



Figure 1 — An enclosure W by H by D holding the heat loads as listed, losing heat through its exposed faces to the ambient, with a fan and the sun on it. Not to scale; the labels are the inputs as entered.

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| loads | Heat loads | 2 row(s) | | spec: Panel schedule PNL-09 | any |
| I_sun | Solar irradiance on the enclosure (0 = indoors) | 800 | W/m² | spec: Site conditions SC-5: 800 W/m² on the sunlit side | 0 … 1200 W/m² |
| A_sun | Sunlit area | 0.48 | m² | assumption: One 600 × 800 face in the sun. | 0 … 100 m² |
| alpha_sun | Surface absorptivity | 0.5 | | assumption: Light-grey paint. | 0.1 … 1 |
| W | Enclosure width | 0.6 | m | spec: Enclosure datasheet AE-1380 | 0.1 … 5 m |
| H | Enclosure height | 0.8 | m | spec: Enclosure datasheet AE-1380 | 0.1 … 5 m |
| D | Enclosure depth | 0.3 | m | spec: Enclosure datasheet AE-1380 | 0.1 … 5 m |
| f_top | Top exposure (1 free, 0 covered) | 1 | | assumption: Free-standing kiosk, bottom on a plinth. | 0 … 1 |
| f_front | Front exposure | 1 | | assumption: Free-standing kiosk, bottom on a plinth. | 0 … 1 |
| f_rear | Rear exposure (0 = wall-mounted) | 1 | | assumption: Free-standing. | 0 … 1 |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| f_left | Left side exposure | 1 | | assumption: Free-standing kiosk, bottom on a plinth. | 0 … 1 |
| f_right | Right side exposure | 1 | | assumption: Free-standing kiosk, bottom on a plinth. | 0 … 1 |
| f_bottom | Bottom exposure (0 = on the floor) | 0 | | assumption: Free-standing kiosk, bottom on a plinth. | 0 … 1 |
| h | Overall heat-transfer coefficient of the enclosure surface (editable reference value) | 5.5 | W/(m²·K) | reference: Reference record: painted steel enclosure, still air, 5.5 W/m²K | 3 … 8 W/(m²·K) |
| T_amb | Ambient temperature | 40 | °C | spec: Site conditions SC-5 | -20 … 55 °C |
| T_max | Maximum internal temperature | 45 | °C | spec: Site conditions SC-5 | 20 … 80 °C |
| altitude | Site altitude (air density) | 0 | m | assumption: Free-standing kiosk, bottom on a plinth. | 0 … 4000 m |
| p_sys | System resistance (filters, grilles) at the reference airflow | 20 | Pa | default | 0 … 500 Pa |
| V_ref | Reference airflow for the system resistance | 100 | m³/h | default | 1 … 100000 m³/h |

| TABLE ROW | VALUES |
|---|---|
| loads[1] | name = PLC and I/O; P = 80 W; qty = 1; duty = 1 |
| loads[2] | name = drives; P = 55 W; qty = 4; duty = 1 |

## 3. Assumptions

A1. Heat leaves the enclosure through its exposed faces at one overall coefficient h (natural convection plus radiation to surroundings at the ambient temperature); covered faces dissipate nothing. (IEC TR 60890-style effective-surface approach; h is a reference value the user may edit.)
A2. Forced ventilation removes the full heat load with air entering at ambient and leaving at the target rise; natural dissipation is neglected for the airflow (conservative). (Manufacturer fan-sizing practice.)
A3. The fan curve is linear between the entered points; the system resistance scales with airflow squared. (Duty-point idealisation.)
A4. A_sun (Sunlit area) is taken as 0.48 m^2. (One 600 × 800 face in the sun.)
A5. alpha_sun (Surface absorptivity) is taken as 0.5. (Light-grey paint.)
A6. f_top (Top exposure (1 free, 0 covered)) is taken as 1. (Free-standing kiosk, bottom on a plinth.)
A7. f_front (Front exposure) is taken as 1. (Free-standing kiosk, bottom on a plinth.)
A8. f_rear (Rear exposure (0 = wall-mounted)) is taken as 1. (Free-standing.)
A9. f_left (Left side exposure) is taken as 1. (Free-standing kiosk, bottom on a plinth.)
A10. f_right (Right side exposure) is taken as 1. (Free-standing kiosk, bottom on a plinth.)
A11. f_bottom (Bottom exposure (0 = on the floor)) is taken as 0. (Free-standing kiosk, bottom on a plinth.)

A12. altitude (Site altitude (air density)) is taken as 0 m. (Free-standing kiosk, bottom on a plinth.)

## 4. Working

[17] P_loads = Σ P_i × n_i × d_i
= 2 item(s)
= 300 W
Internal heat load

[18] P_solar = alpha_sun * I_sun * A_sun
= 0.5 × 800 W/m² × 0.48 m²
= 192 W
Solar gain

[19] P_total = P_loads + P_solar
= 300 W + 192 W
= 492 W
Total heat to remove

[20] A_eff = f_top * W * D + f_bottom * W * D + f_front * W * H + f_rear * W * H + f_left * H * D [1](approach)
= 1 × 0.6 m × 0.3 m + 0 × 0.6 m × 0.3 m + 1 × 0.6 m × 0.8 m + 1 × 0.6 m × 0.8 m + 1 × 0.8 m × 0.8 m × 0.3 m
= 1.62 m²
Effective dissipating surface

[21] dT_nat = P_total / (h * A_eff)
= 492 W / (5.5 W/(m²·K) × 1.62 m²)
= 55.219 K
Temperature rise with natural dissipation

[22] T_int = T_amb + dT_nat
= 40 degC + 55.219 K
= 95.219 degC
Internal temperature with natural dissipation

[23] dT_allow = T_max - T_amb
= 45 degC − 40 degC
= 5 K
Allowed rise

[24] u_nat = dT_nat / dT_allow
= 55.219 K / 5 K
= 11.044
Natural dissipation utilisation

[25] P_nat_max = h * A_eff * dT_allow
= 5.5 W/(m²·K) × 1.62 m² × 5 K
= 44.55 W
Heat the enclosure dissipates naturally at the limit

[26] P_cool = max(0 W, P_total - P_nat_max)
= max(0 W, 492 W − 44.55 W)
= 447.45 W
Cooling-unit capacity needed if ventilation is not used

[27] p_site = 101325 (1 - 2.25577e-5 z)^5.2559
= z = 0 m
= 101.32 kPa
Site pressure

[28] rho_air = p_site / (287.055 J/(kg*K) * T_amb)
= 101.32 kPa / (287.055 J/(kg*K) × 40 degC)
= 1.1272 kg/m³
Air density at ambient and altitude

[29] V_req = P_total / (rho_air * 1006 J/(kg*K) * dT_allow)
= 492 W / (1.1272 kg/m³ × 1006 J/(kg*K) × 5 K)
= 312.39 m³/h
Airflow for forced ventilation at the allowed rise (natural dissipation neglected)
[30] V_req_net = max(0 m^3/h, (P_total - P_nat_max) / (rho_air * 1006 J/(kg*K) * dT_allow))
= max(0 m^3/h, (492 W - 44.55 W) / (1.1272 kg/m³ × 1006 J/(kg*K) × 5 K))
= 284.11 m³/h
Airflow crediting natural dissipation

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| P_loads | 300 W | Internal heat load |
| P_solar | 192 W | Solar gain |
| P_total | 492 W | Total heat |
| A_eff | 1.62 m² | Effective surface |
| dT_nat | 55.219 K | Natural temperature rise |
| T_int | 95.219 degC | Internal temperature (natural) |
| P_nat_max | 44.55 W | Natural dissipation at the limit |
| P_cool | 447.45 W | Cooling unit needed |
| rho_air | 1.1272 kg/m³ | Air density |
| V_req | 312.39 m³/h | Required airflow |
| V_req_net | 284.11 m³/h | Required airflow (net) |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| FAIL | Natural dissipation holds T_max | u_nat = 11.044 | u_nat ≤ 1 | 11 | |

## 7. Applicability

WARNING — input.p_sys: p_sys (System resistance (filters, grilles) at the reference airflow) was not entered; the template default 20 Pa is used.
WARNING — input.V_ref: V_ref (Reference airflow for the system resistance) was not entered; the template default 100 m^3/h is used.
WARNING — cooling-unit: T_max: natural dissipation holds 44.55 W of 492 W; ventilate or fit a 447.4 W cooling unit.

## 8. Verification

Template elec.enclosure-heat@1.0.1 Content hash 54155df64960dd90 Engine calc-engine/1.0.0
Known-answer cases passed 2/2 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

## 9. References

1. International Electrotechnical Commission. A Method of Temperature-Rise Verification of Low-Voltage Switchgear and Controlgear Assemblies by Calculation. IEC TR 60890. IEC.
2. Rittal; nVent Hoffman. Enclosure climate control guides [manufacturer technical literature].
3. British Standards Institution. Low-Voltage Switchgear and Controlgear Assemblies—Part 1: General Rules. BS EN 61439-1. BSI.